AREL URBANOZO

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EDUCATION

University of Alabama in Huntsville — Huntsville, AL

Bachelor of Science in Aerospace Engineering — *Expected May 2027*

- GPA: 3.85/4.00
- Honors: President's List (F23, S24) · Dean's List (F24, S25, F25) · UAH Merit Scholarship

Relevant Coursework

- Spacecraft & Mission Design: MAE 490 Mission Design (capstone: Radiation Test Bed IPT); MAE 469 Introduction to Astrodynamics; MAE 200 Principles of Aeronautics & Astronautics
- Propulsion & Flight: MAE 540 Rocket Propulsion (graduate); MAE 580 Aircraft Stability & Control (graduate); MAE 330/331 Aerodynamics & Lab; MAE 343 Compressible Aerodynamics; MAE 341 Thermodynamics I
- Controls, Instrumentation & Electronics: MAE 488 Systems & Controls; MAE 311 Measurement & Instrumentation; EE 213 Electrical Circuits & Systems; EE 315 Electronic Analysis & Design
- Structures & Mechanics: MAE 271 Statics; MAE 272 Dynamics; MAE 370/375 Mechanics of Materials & Lab; MAE 371 Aerospace Structures
- Computation & Mathematics: MAE 211 Computational Tools; MAE 284 Numerical Methods; MA 238 Differential Equations; MA 244 Linear Algebra; EGR 101 Computing for Engineers
- Management & Mentoring: ISE 321 Engineering Economy; EGR 299/399 Engineering Mentoring I & II

RESEARCH & ENGINEERING EXPERIENCE

UAH IDEAS Lab — ChargerSat-2 — Huntsville, AL

Lead Systems Engineer & Communications Lead (Research Assistant) — *Feb 2026 – Present*

- Lead systems engineering for a 12U NASA Class D CubeSat (on-orbit autonomous-AI demo), coordinating interfaces, requirements decomposition, and verification-matrix closure across 8 subsystem teams through the Phase A MDR/SRR gate.
- Authored the communications plan and link-margin analysis; closed a 50 Mbps X-band LEO downlink at +4.04 dB and showed the 100 Mbps option (+2.03 dB) fails the 3 dB requirement.
- Identified that the baselined X-band radio cannot reach the 8400–8450 MHz DSN deep-space band, driving an S-band cislunar re-plan and FCC Part 5 Experimental licensing as the primary regulatory path.
- Authored the Universal Payload ICD (≈6.5U, ≤8.5 kg, ≤42 W orbit-average); built an RF Mission Workbench (Python MILP scheduler, Three.js globe) for downlink planning.
- Established a controlled Phase A document set — SEMP, communications plan, regulatory baseline, Class D 5×5 risk register, configuration management plan — with automated requirements-traceability checks.
- Present design and trade results at internal milestone reviews to the Chief Engineer and program leadership.

UAH IDEAS Lab — INTREPID (NASA-funded lunar CubeSat outreach) — Huntsville, AL

Research Assistant — Curriculum, Hardware & Test — *Feb 2026 – Present*

- Authored 17-week Sensor and Testing Team curricula (guides, teacher notes, decks) for high-school CubeSat teams.
- Designed the student payload PCB in KiCad (95×95 mm, 2-layer; RP2040, MPU-6050 IMU, thermistor) from schematic to fab-ready Gerbers in one week.
- Led a deployer velocity test campaign (n = 7, 997 Hz IMU + photogate); proved the IMU captured only 73% of the true 1.08 m/s separation and traced the error to fixture impact (pulse 5.7× narrower than spring theory), ruling out mount resonance.
- Deliver on-site instruction and mentorship at partner high schools; coordinate logistics and test days.
- Built offline classroom tools that bypass school web filters: a Hooke's Law / CubeSat deployment simulator (React/TypeScript) and CircuitPython Studio, a Chrome extension with Web Serial REPL and live plotting.

TEACHING & MENTORING

UAH College of Engineering — Huntsville, AL

Teaching Assistant — EGR 101 Intro to Computing for Engineers — *Aug 2025 – May 2026*

- Supported a 20-student section in lecture and office hours, debugging Python/MATLAB; graded ~40 assignments weekly.
- Served two consecutive semesters under two instructors; coached students on problem decomposition and assignment planning.

INTREPID High-School CubeSat Program — Facilitator & Curriculum Author — 2026 – Present

- Designed and teach 17 weeks of hands-on CubeSat instruction for grades 10–12 at partner schools (see Research Experience).

UAH College of Engineering — Engineering Mentoring I & II (EGR 299/399) Fall 2024, Fall 2025

STUDENT LEADERSHIP

UAH Space Hardware Club — CanSat 2024, Team 2078 "Shockwave" Huntsville, AL
Team Lead *Oct 2023 – Jun 2024*

- Led a 10-person team as a freshman through PDR, CDR, environmental test, flight, and post-flight review; placed 2nd internationally and top U.S. team (score 99.52) at the AAS CanSat Competition.
- Ran component and design trade studies; designed and iterated the flight PCB seven times in KiCad (Teensy 4.1, BMP390, BNO055, XBee telemetry, 3 regulators).

UAH Space Hardware Club — Ho'omau Two-Month CanSat Huntsville, AL
Electrical / PCB Designer *Aug 2023 – Nov 2023*

- Designed CanSat avionics (Raspberry Pi Pico, BMP390, LSM6DSOX, XBee, MOSFET-driven release actuation) through three schematic revisions and a 2-layer PCB.

TECHNICAL PROJECTS

Hooke's Law & CubeSat Deployment Simulator Aug 2026
React, TypeScript, Vite

- Offline physics lab for Chromebooks: Verlet-integrated CubeSat deployer challenge (1U–6U, 1.2–1.8 m/s, ≤15 g), WCAG 2.1 AA.

PMI Dynamometer Apr 2026
KiCad, C++, MATLAB

- Torque/RPM dynamometer with custom PCB (Pico 2, HX711 load cell, Hall-effect RPM), MATLAB GUI, and load-cell uncertainty analysis.

CircuitPython Studio Aug 2026
JavaScript, Web Serial API

- Manifest V3 Chrome extension: CircuitPython editor, serial REPL, CIRCUITPY drive sync, and 60 FPS live plotter with CSV export — zero network requests.

Interplanetary Transfer Analysis (MAE 469) Spring 2026
MATLAB

- Lambert-based porkchop plots for Earth–Mars and Earth–Jupiter transfer windows.

Aircraft Sizing & Performance Fall 2024
Python, SciPy

- Conceptual sizing and performance analysis for Denver–Huntsville and Honolulu–Boston missions (github.com/urbznspices/Aero).

TECHNICAL DOCUMENTS AUTHORED

- Communications Plan & Link Margin Analysis — ChargerSat-2 (2026)
- Communications Licensing & Regulatory Research Baseline — ChargerSat-2 (2026)
- Universal Payload Interface Control Document, Draft 0.2 — ChargerSat-2 (2026)
- Systems Engineering Management Plan, Rev 4.x — ChargerSat-2 (Lead Systems Engineer, 2026)
- Deployment Velocity Test Report, Runs 006–012 — INTREPID (Aug 2026)
- INTREPID Sensor Team & Testing Team Student Guides, Weeks 1–17 (2026)
- CanSat 2024 PDR, CDR, Environmental Test & Post-Flight Review packages — Team 2078 Shockwave (contributor & lead, 2024)

HONORS & AWARDS

- 2nd Place International, Top U.S. Team — AAS CanSat Competition 2024 (Team Lead, "Shockwave")
- President's List — Fall 2023, Spring 2024
- Dean's List — Fall 2024, Spring 2025, Fall 2025
- UAH Merit Scholarship — 2023

OTHER EMPLOYMENT

UAH Enhanced Teaching & Learning Center Huntsville, AL
Student Specialist *Oct 2024 – May 2025*

- Diagnosed A/V failures across 130 technology-enabled classrooms; installed projectors, cameras, and rack hardware.

SKILLS

- Systems Engineering: NASA SE lifecycle, requirements & VCRM, ICDs, trade studies, risk & config. management, design reviews
- RF & Mission: Link budgets, TT&C, ground-contact planning, spectrum licensing, power & duty-cycle budgets, astrodynamics
- Hardware & Test: PCB design (KiCad, 5+ boards), embedded sensors, TH/SMD soldering, instrumentation & DAQ, test & analysis
- Software: Python (NumPy, SciPy), MATLAB, C/C++, CircuitPython, TypeScript/React, Svelte, Git, Inventor, Onshape, LTspice

BEYOND ENGINEERING

- Published classical music composer; writing original musicals (book, music & lyrics)
- UAH Rowing Club
- Raised in Kamuela on the Island of Hawai'i